

Assignment - 01

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Fit a curve at the form $y = ax + b$ for the following data.

x	y
2	2.4
3	3
4	3.6
5	4
6	5

<u>x</u>	<u>y</u>	<u>xy</u>	<u>x²</u>
1	2.4	2.4	1
2	3	6	4
3	3.6	10.8	9
4	4	16	16
5	5	25	25
6	6	36	36
<u>Σ</u>	<u>24</u>	<u>26.8</u>	<u>91</u>

$$n = 6$$

$$y = ax + b$$

$$\Sigma y = a \Sigma x + nb$$

$$\Sigma xy = a \Sigma x^2 + b \Sigma x$$

$$(24 = 21a + 6b) \rightarrow \text{①}$$

$$(96.2 = 91a + 216b) \rightarrow \text{②}$$

$$16.8 = 91a + 216b$$

$$-168 \Rightarrow 147a + 42b$$

$$+ 192.4 \Rightarrow 182a + 42b$$

$$24.4 = 35a$$

$$a = \frac{24.4}{35} = 0.69714$$

$$24 = 21(0.69714) + 6b$$

$$b = 1.56$$

$$y = (0.69714)x + 1.56$$

2) Fit a curve of the form $y = ab^x$ for the following data:

x	0	2	4	5	7	10
y	100	120	256	390	710	1600

$$y = ab^x$$

$$\log y = \log a + x \log b$$

$$Y = A + xB$$

$$\sum Y = nA + B \sum x$$

$$n = 6$$

$$\sum xY = A \sum x + B \sum x^2$$

<u>x</u>	<u>y</u>	<u>Y = log y</u>	<u>xx</u>	<u>x^2</u>
0	100	1.60514	0	0
2	120	1.78719	9.57198	4
4	256	5.51518	22.18042	16
5	390	5.98615	29.93225	25
7	710	6.56526	15.95682	49
10	1600	7.37776	78.7776	100
<u>$\sum 28$</u>	<u>$\sum = 34.86731$</u>		<u>$\sum 181.42227$</u>	<u>$\sum 194$</u>

$$(34.86731 = 6A + 28B) \times 28$$

$$(181.42227 = 28A + 194B) \times 6$$

$$- 976.28468 = 168A + 784B$$

$$+ 1088.53422 = 168A + 1164B$$

$$112.24954 = 380B$$

$$B = 0.29539$$

$$34.86731 = 28(0.29539) + 6A$$

$$A = 1.13272$$

$$\log a = A \Rightarrow a = e^A$$

$$a = e^{4.43272} \Rightarrow 84.16002$$

$$\log b \Rightarrow b = e^B$$

$$b = e^{0.29539} \Rightarrow 1.34365$$

$$y = (84.16002) \cdot (1.34365)^x$$

3) Calculate the coefficient of correlation and obtain the lines of regression for the following data.

x	1	2	3	4	5	6	7	8	9
y	9	8	10	12	11	13	14	16	15

$$n = 9 \quad \bar{x} = \frac{\sum x}{n} = \frac{45}{9} = 5$$

$$\bar{y} = \frac{108}{9} = 12$$

$$r = \frac{\sum xy}{\sqrt{\sum x^2} \sqrt{\sum y^2}}$$

<u>x</u>	<u>X = x - \bar{x}</u>	<u>y</u>	<u>Y = y - \bar{y}</u>	<u>X²</u>	<u>Y²</u>	<u>XY</u>
1	-4	9	-3	16	9	12
2	-3	8	-4	9	16	12
3	-2	10	-2	4	4	4
4	-1	12	0	1	0	0
5	0	11	-1	0	1	0
6	1	13	1	1	1	1
7	2	14	2	4	4	4
8	3	16	4	9	16	12
9	4	15	3	16	9	12

$$= 0.0000045 \quad (0.98)^{494}$$

$$= 0.0000045$$

$$1 - 0.0000045$$

$$= 0.999995$$

(ii) $1 - {}^{500}C_0 (0.02)^0 (0.98)^{500}$
 $= 1 - 0.000015$
 $= 0.999985$

(iii) $P(x) = P(2) + P(3) + P(4)$
 $= {}^{500}C_2 (0.02)^2 (0.98)^{498} + {}^{500}C_3 (0.02)^3 (0.98)^{497} + {}^{500}C_4 (0.02)^4 (0.98)^{496}$
 $= 0.0023 + 0.0074 + 0.0189$
 $= \underline{\underline{0.0285}}$

(iv) ${}^{500}C_{15} (0.02)^{15} (0.98)^{485}$
 $= \underline{\underline{0.0347}}$

→ A Certain Screw making machine produces on an average two defectives out of 100 and places them in boxes of 500. find the probability that the box contains.

→ Three defectives
 → At least one defective

Expected defectives per box,
 $= \frac{2}{100} \times 500 = 10$

$P = \frac{2}{100} = 0.02$

$\sqrt{q = 0.98}$

$\sqrt{P = 0.02}$

$\sqrt{N = 500}$

$$\begin{aligned}
 \text{(iii)} \quad N &= {}^nC_x \cdot p^x \cdot q^{n-x} \\
 &= 800 \left[{}^5C_3 \left(\frac{1}{2} \right)^3 \left(\frac{1}{2} \right)^2 \right] \\
 &= 800 (1) \left(\frac{1}{2^5} \right) \\
 &= \frac{800}{2^5} \\
 &= \underline{\underline{25 \text{ families}}}
 \end{aligned}$$

7) In 800 families of 5 children each how many families would be expected to have:

- i) 3 boys 2 girls
- ii) 2 boys and 3 girls
- iii) no girls

iv) At most 2 girls by assuming for boys and girls to be equal.

Let $P = \text{boys}$ $q = \text{girls}$
assuming to be equal.

$$\Rightarrow P + q = 1 \quad \Rightarrow P = \frac{1}{2} \quad \& \quad q = \frac{1}{2}$$

Considering x as number of boys.

$$\begin{aligned}
 \text{v)} \quad N &= {}^nC_x \cdot p^x \cdot q^{n-x} \\
 &= 800 ({}^5C_3 \cdot P^3 \cdot q^2) \\
 &= 800 \left({}^5C_3 \cdot \left(\frac{1}{2} \right)^3 \cdot \left(\frac{1}{2} \right)^2 \right)
 \end{aligned}$$

$$= 800 \times \frac{5 \times 4 \times 3}{2 \times 1} \times \frac{1}{2^5} \Rightarrow \underline{\underline{250 \text{ families}}}$$

(i) $P(x < 4)$

$$= 0.02 + 0.06 + 0.14 + 0.14$$

$$= \underline{0.32}$$

(ii) $P(x > 5)$

$$= \underline{0.26}$$

(iii) $P(3 < x < 6)$

$$= 0.18 + 0.22$$

$$= \underline{0.4}$$

6) Find the value of C such that $f(x) = \begin{cases} 0 < x \end{cases}$
a Probability density function.

$$C \int_a^b f(x) dx = 1$$

$$C \int_0^3 x^2 dx = 1$$

$$C \left[\frac{(3)^2}{3} - \frac{(0)^3}{3} \right] = 1$$

$$9C = 1$$

$$C = \frac{1}{9}$$

$$N({}^nC_r \cdot p^r \cdot q^{n-r})$$

$$= 800 ({}^5C_2 \cdot p^2 \cdot q^3)$$

$$= 800 ({}^5C_2 \cdot (1/2)^2 \cdot (1/2)^3)$$

$$= 800 \times \frac{5 \times 4}{2 \times 1} \times \frac{1}{8}$$

$$= \underline{250 \text{ families}}$$

The regression line of y on x is

$$(y - \bar{y}) = \frac{\sum xy}{\sum x^2} (x - \bar{x}) \Rightarrow (y - 15) = (1.761471) (x - 9) \rightarrow ①$$

$$y = 1.761471x + 2.64705 \rightarrow ①$$

The regression line of x on y

$$(x - \bar{x}) = \frac{\sum xy}{\sum y^2} (y - \bar{y}) \Rightarrow (x - 9) = (0.1401) (y - 15) \rightarrow ②$$

$$x = 0.1401y + 0.3985 \rightarrow ②$$

from ① and ②

$$r = \sqrt{b_{xy} * b_{yx}} = \sqrt{0.1401 * 1.76147}$$

$$\Rightarrow \boxed{r = 0.58127}$$

The Probability density function of a variable x is

x	0	1	2	3	4	5	6
$P(x)$	K	$3K$	$5K$	$7K$	$9K$	$11K$	$13K$

Find (i) $P(x < 4)$ (ii) $P(x > 5)$ (iii) $P(3 < x < 6)$

$$49K = 1$$

$$\Rightarrow K = \frac{1}{49} \Rightarrow \boxed{K = 0.02}$$

x	0	1	2	3	4	5	6
$P(x)$	0.02	0.06	0.1	0.14	0.18	0.22	0.26

$$(0.02 + 0.06 + 0.1 + 0.14 + 0.18 + 0.22 + 0.26) = 1$$

$$① P(x < 4) = 0.02 + 0.06 + 0.1 + 0.14 = 0.32$$

4) Obtain the lines of regression and hence find the co-efficient of correlation for following data

x	1	3	4	2	5	8	9	10	13	15
y	8	6	10	8	12	16	16	10	32	32

$$\bar{x} = \frac{\sum x}{n} = \frac{70}{10} = 7 \quad \bar{y} = \frac{\sum y}{n} = \frac{150}{10} = 15$$

x	y	$x - \bar{x}$	$y - \bar{y}$	x^2	y^2	xy
1	8	-6	-7	36	49	12
3	6	-4	-9	16	81	26
4	10	-3	-5	9	25	15
2	8	-5	-7	25	49	35
5	12	-2	-3	4	9	6
8	16	1	1	1	1	1
9	16	2	1	4	1	1
10	10	3	-5	9	25	-15
13	32	6	17	36	289	102
15	32	8	17	64	289	136
				<u>204</u>	<u>818</u>	

$$b_{xy} = \frac{\sum xy}{\sum y^2} = \frac{360}{818} = 0.4401$$

$$b_{yx} = \frac{\sum xy}{\sum x^2} = \frac{360}{204} = 1.7647$$

The regression line of y on x

$$(y - \bar{y}) = \frac{\sum xy}{\sum x^2} (x - \bar{x})$$

$$(y - 15) = (1.7647) (x - 7)$$

$$y = (1.7647)x + 2.64703 \rightarrow \text{①}$$

$$\sum xy = 57 \quad \sum x^2 = 60, \quad \sum y^2 = 60$$

$$r = \frac{\sum xy}{\sqrt{\sum x^2} \sqrt{\sum y^2}} = \frac{57}{\sqrt{60} \sqrt{60}} = \frac{57}{60} = 0.95$$

$$\sqrt{x^2} = \frac{\sum x^2}{n} = \frac{60}{9} = 6.67 = \boxed{x = \sqrt{6.67}}$$

$$\sqrt{y^2} = \frac{\sum y^2}{n} = \frac{60}{9} = 6.67 = \underline{\underline{\sqrt{6.67}}}$$

$$\sqrt{x} = 2.58263, \quad \sqrt{y} = 2.58263$$

The regression line of y on x is

$$(y - \bar{y}) = \frac{r \sigma_y}{\sigma_x} (x - \bar{x})$$

$$(y - 12) = \frac{(0.95)(2.58)}{2.58} (x - 5)$$

$$\boxed{y = 0.95x + 7.25}$$

The regression of x on y is

$$(x - \bar{x}) = \frac{r \sigma_x}{\sigma_y} (y - \bar{y})$$

$$(x - 5) = \frac{0.95(2.85)}{2.85} (y - 12)$$

$$\boxed{x = 0.95y - 6.1}$$